

Explainable autoencoder for neutron star dense matter parameter estimation

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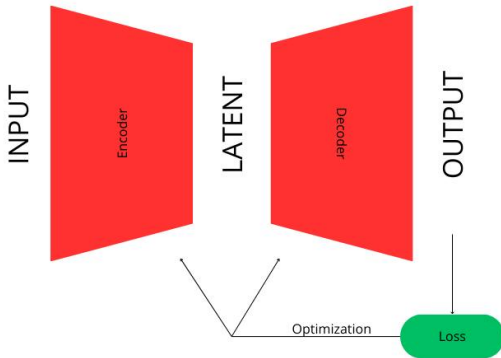
- ① Introduction
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Introduction: AutoEncoder (AE)

- Ideal case:

$$\begin{aligned} \text{output} &= \text{input} \\ & (= \text{Decoder}[\text{Encoder}(\text{input})]) \end{aligned}$$

- Unsupervised: while basic AEs [1] are usually seen as black boxes, a huge effort is now done explain the latent space (explainability = why it works like this?).
- **Machine Learning needs explainability** to link data to physical laws.



- Tolman-Oppenheimer-Volkoff (TOV) equations [2]:

$$\begin{cases} \frac{dP}{dr} = -\frac{Gm}{r^2} \rho \left(1 + \frac{P}{\rho c}\right) \left(1 + \frac{4\pi r^3 P}{mc^2}\right) \left(1 - \frac{2Gm}{rc^2}\right)^{-1} \\ \frac{dm}{dr} = 4\pi r^2 \rho \end{cases} .$$

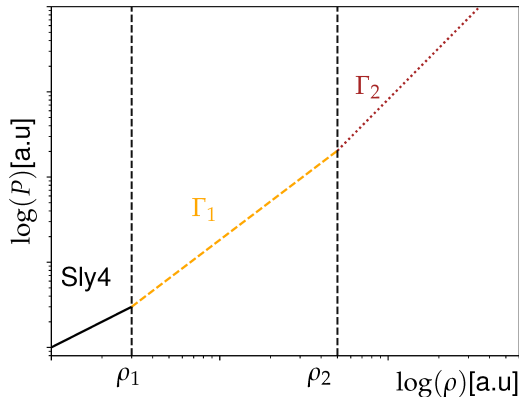
- Piecewise-polytropic EoS:

$$P = \begin{cases} K\rho^{\gamma_1} & \text{if } \rho < \rho_1 \\ K\rho^{\gamma_2} & \text{if } \rho_1 < \rho < \rho_2 \\ K\rho^{\gamma_3} & \text{if } \rho > \rho_3 \end{cases} .$$

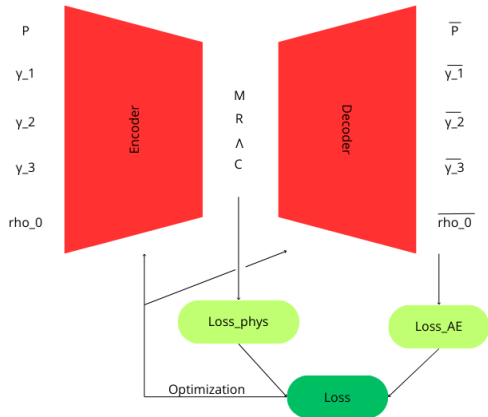
Introduction: Neutron Stars' (NS) Equation of State (EoS)

- Once specified the EoS and the central density ρ_c , we know how to numerically solve TOV equations to get mass M , radius R , tidal deformability Λ and compactness $C = M/R$ values.
- **The inverse is not straightforward:** for one couple of (M, R) values more than one EoS can be valid.
- Machine Learning techniques could be used to approximate the inverse function, **but we must have explainability.**

- Considering piecewise-polytropic EoSs, **we selected $1.4 \cdot 10^4$ sets of values for $P_1, \gamma_1, \gamma_2, \gamma_3, \rho_0$** , uniformly varied within intervals chosen to produce quantities consistent with theoretical and observational constraints.
- For each of them, we numerically solved the TOV equations to have (M, R, Λ, C) **target values**.
- Then, we randomly selected $5 \cdot 10^3$ samples in the whole dataset to actually train the network.



Architecture: Explainable AE



$$\mathcal{L} = \mathcal{L}_{AE} + \alpha \mathcal{L}_{TOV} + \beta \mathcal{L}_C + \iota \mathcal{L}_{\Lambda C}$$

$$\mathcal{L}_{AE} = \|(P_1, \gamma_1, \gamma_2, \gamma_3, \rho_0)_{\text{input}} - (P_1, \gamma_1, \gamma_2, \gamma_3, \rho_0)_{\text{output}}\|^2$$

$$\mathcal{L}_{TOV} = \|(M, R, \Lambda, C)_{\text{latent}} - (M, R, \Lambda, C)_{\text{target}}\|^2$$

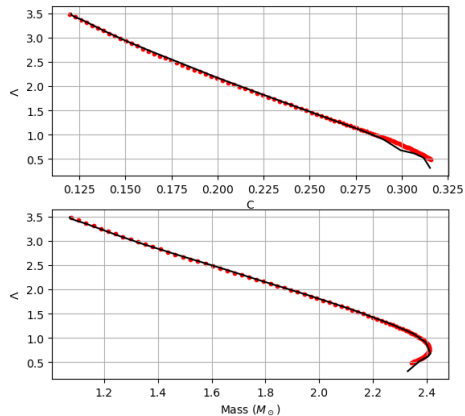
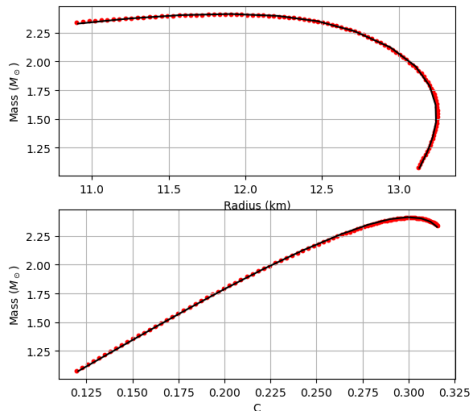
$$\mathcal{L}_C = \left\| \frac{M_{\text{latent}}}{R_{\text{latent}}} - C_{\text{target}} \right\|^2$$

$$\mathcal{L}_{\Lambda C} = \left\| \Lambda_{\text{latent}} - f\left(\frac{M_{\text{latent}}}{R_{\text{latent}}}\right) \right\|^2$$

$$f \text{ is the inverse of } C = \sum_i b_i (\log(\Lambda))^i \quad [3]$$

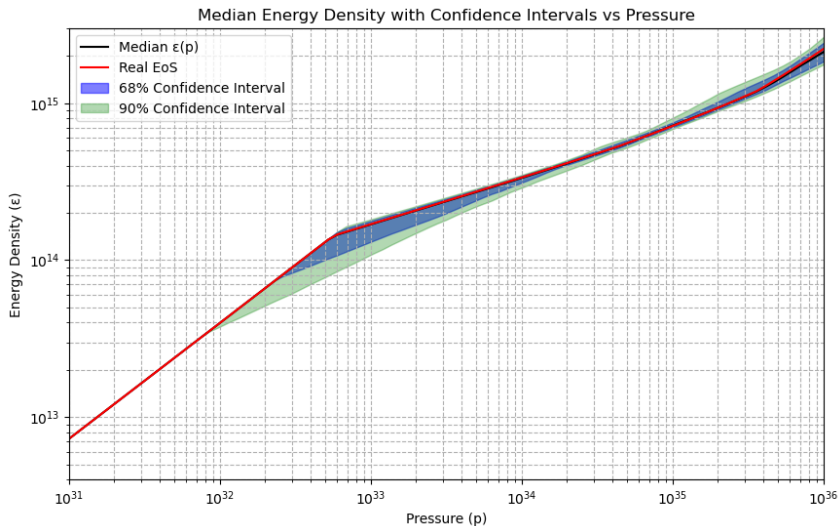
Results: Trained encoder's test

- **Input:** DD2's parameters ($P_1 = 10^{34.638}$ dyn/cm², $\gamma_1 = 3.414$, $\gamma_2 = 3.097$, $\gamma_3 = 2.322$, $\rho_1 = 3.1 \cdot 10^{14}$ g/cm², $\rho_2 = 2 \cdot 10^{15}$ g/cm² [4]).
- **Black line:** numerical TOV solution (M, R, Λ, C)
- **Red dots:** encoded M, R, Λ, C

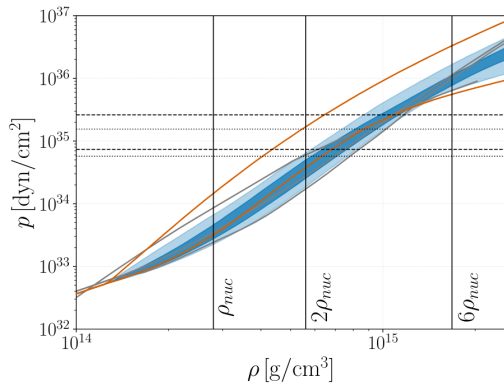
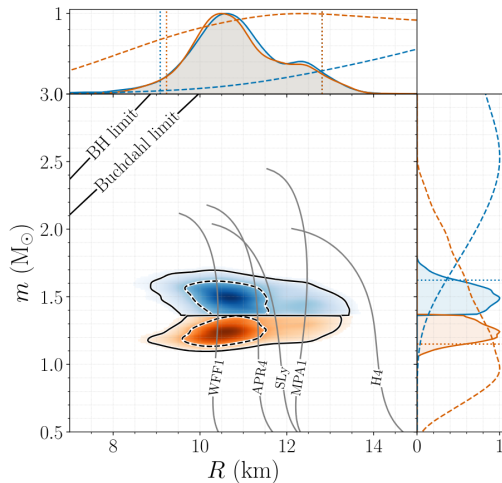


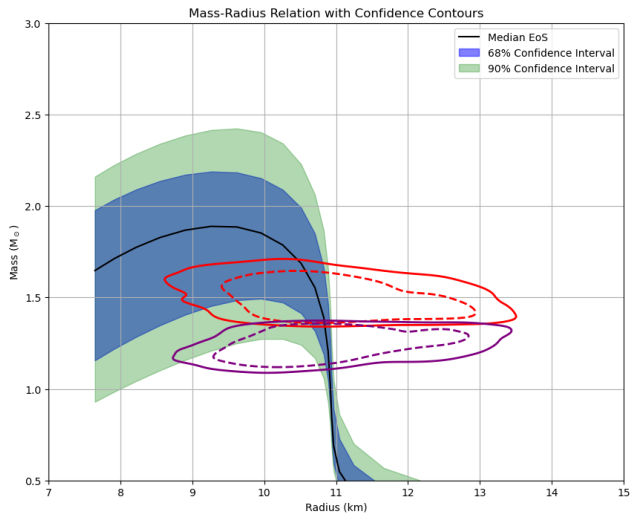
Results: Trained decoder's test

- **Input:** numerical TOV solution (M, R, Λ, C ; previous black line)

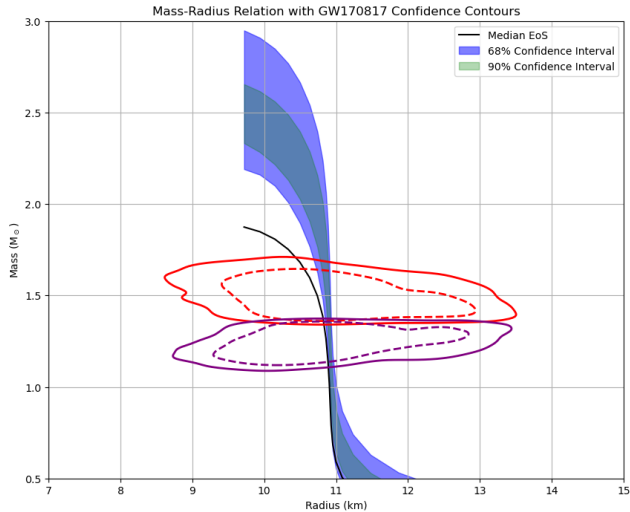


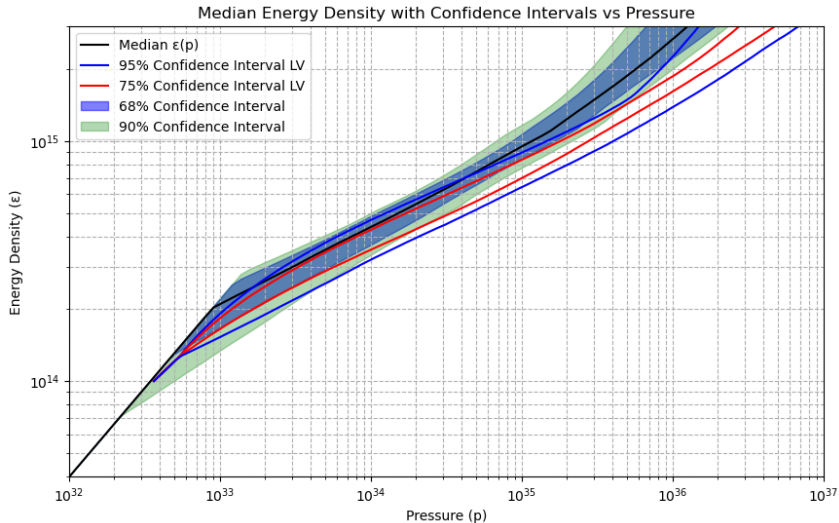
- LVK collaboration [5] produced Bayesian inference on GW170817 data, providing piecewise-polytropic parameter's values and Mass-Radius samples.
- As a test for our algorithm (the decoder part), we compared our prediction for the EoS with the LVK collaboration one.





GW170817: Results - Mass-Radius





- Our autoencoder is able to solve the TOV equation and to generate a precise reconstruction of the EoS.
- The physical constraint on the latent space permits to have explainability of the network.
- We applied the trained decoder to real GW data, recovering an EoS consistent with LVK collaboration results.

- [1] D. E. Rumelhart, G. E. Hinton, and R. J. Williams, "Learning internal representations by error propagation," 1986.
- [2] J. R. Oppenheimer and G. M. Volkoff, "On massive neutron cores," *Phys. Rev.*, vol. 55, pp. 374–381, Feb 1939.
- [3] S. Shashank, F. Hossein Nouri, and A. Gupta, "f -mode oscillations of compact stars with realistic equations of state in dynamical spacetime," *New Astronomy*, vol. 104, p. 102067, 06 2023.
- [4] B. Kumar and P. Landry, "Inferring neutron star properties from gw170817 with universal relations," *Physical Review D*, vol. 99, June 2019.
- [5] B. Abbott *et al.*, "Gw170817: Measurements of neutron star radii and equation of state," *Physical Review Letters*, vol. 121, Oct. 2018.